Internal division of a segment

1 When the point P divides the $\overline{AB}$ internally, is the ratio $m : n$ positive or negative?

2 Point P divides the interval joining the points (x_1, y_1) and (x_2, y_2) in the ratio $m : n$.

a State the formula which gives the x-coordinate of point P.

b State the formula which gives the y-coordinate of point P.

3 $\overline{AB}$ is divided internally in the ratio $5 : 3$ by a point P, where A and B are the points $(1, 13)$ and $(3, 11)$ respectively.
Find the coordinates of point P.

4 For each of the following, find the coordinates of point P:

a P divides the interval joining points $(5, 10)$ and $(1, 2)$ in the ratio 3:1.

b P divides the interval joining points $(2, 9)$ and $(-1, 3)$ in the ratio 2:1.

c P divides the interval joining points $(-1, 10)$ and $(1, 11)$ in the ratio 4:3.

5 $\triangle ABC$ has vertices at $A\,(4, -4)$, $B\,(5, 6)$ and $C\,(3, 4)$.

a Find the coordinates of P, the point which divides $\overline{AB}$ in the ratio $4 : 1$.

b Find the length of $\overline{CP}$.

6 $\triangle ABC$ has vertices at $A\,(3, 4)$, $B\,(5, 2)$ and $C\,(7, 7)$.

a Find the coordinates of point X, the point which divides $\overline{AB}$ in the ratio $5 : 1$.

b Find the coordinates of point Y, the point which divides $\overline{AC}$ in the ratio $5 : 1$.

c Find the slope of $\overline{BC}$.

d Find the slope of $\overline{XY}$.

e Is $\overline{XY}$ parallel to $\overline{BC}$?

External division of a segment

7 When the point P divides $\overline{AB}$ externally, is the ratio $m : n$ positive or negative?

8 For each of the following, find the coordina[redacted] point P:

a $\overline{AB}$ is divided externally in the ratio $-4:3$ by a point $P(x,y)$, where A and B are the points $(2,9)$ and $(3,12)$ respectively.

b $\overline{AB}$ is divided externally in the ratio $-5:1$ by a point $P(x,y)$, where A and B are the points $(5,7)$ and $(3,1)$ respectively.

9 Point $P(x,y)$ divides the interval joining points $A(-8,5)$ and $B(-5,6)$ in the ratio $-3:2$.
Find the coordinates of point P.

10 Point $P\left(1,\dfrac{61}{6}\right)$ divides the interval between the points $A(2,11)$ and $B(-4,6)$ in the ratio $m:n$.

a What ratio does P divide $\overline{AB}$ into?

b What ratio does B divide $\overline{AP}$ into?

c What ratio does A divide $\overline{PB}$ into?

11 A line segment has endpoints $S(-3,8)$ and $T(12,-4)$. Point R lies on $\overline{ST}$ such that the ratio of SR to RT is $1:2$.
Find the coordinates of point R.